

Thibaut Cuvelier

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Publons: <https://publons.com/researcher/3491439/thibaut-cuvelier/>

Applications of operations research (decision science) to industrial issues

Research keywords: mathematical optimisation / operational research, reinforcement learning

Research experience

Since September 2021: **software engineer** at Google Research (Operations Research group).

I am working on topics related to middle-mile and last-mile **logistics**. My team is responsible for the development of the solver technology behind [Google Cloud Fleet Routing API](#) and [Google Maps Platform Route Optimization](#), a large part of which is open-sourced as the C++ library [OR-Tools](#).

Regarding **middle-mile logistics**, I define a new product and research state-of-the-art technologies, based on ideas from reinforcement learning, to implement a low-latency, incremental solver for a problem that is intractable with current MIP approaches. Our current approach is based on a Dantzig-Wolfe-like **decomposition** of the problem without an explicit MIP/LP formulation to increase computational efficiency. I implement the **graph algorithms** and the distributed infrastructure for **production** in C++.

These graph algorithms are now also used within Google's network-routing infrastructure, helping **decrease the computation times by 4x**. This work has been awarded a Perfy award.

Related publications: [W3], [T7], [A10], [A12].

Within Google Maps Platform Route Optimization, I refactored existing C++ code to improve the user experience of API users through better error reporting and to lower tech debt. I was responsible for a large-scale internal refactoring that I pushed to production without causing downtime.

I also use **large language models** (LLMs) to help onboard new users in the framework of **agentic AI**: I am collaborating on a set of agents that provide a new interface for the API using Google ADK, including their evaluation.

Related publications: [P5].

I also implemented new features in an existing C++ solver for **vector bin packing**, integrating new constraints into column generation. My work has enabled some simulations to be **sped up by 30x**.

Google Research profile: <https://research.google/people/ThibautCuvelier/>

Major community contributions:

- Hosted Onno Eberhard as an **intern** (June-September 2023) on applications of reinforcement learning to middle-mile logistics. Related publications: [W3], [T8], [A10].
- Co-hosted Aymane Lotfi as a **student researcher** (January-June 2024) on applications of reinforcement learning to combinatorial optimisation (set

covering) and on instance generation for middle-mile logistics. Related publications: [T8], [A11].

- Co-hosted Achref Jaziri as a **student researcher** (July-December 2025) on applications of machine learning to combinatorial optimisation (set covering). Related publication: [C4].
- Driving force behind the [Operations Research resources](#).
- OR-Tools maintainer. I have improved the documentation, reworked the graph API (I wrote the code for k shortest paths) and routing solver, and released parsers. I am working on a Julia wrapper for the generic optimisation solvers and mentoring Warren Ochibobo on the project. I ported the suite to the MinGW compiler. Related publications: [A12], [A13], [A14], [T9].

Other highlights:

- Fully trained interviewer since July 2022. Conducted 35+ coding interviews.
- **Readability:** [API](#) (since October 2022), C++ (since January 2023), and Python (since April 2026).

October 2017-June 2021: **doctoral researcher** at Orange Labs (MORE department) and CentraleSupélec (Paris-Saclay university).

Keywords: mathematical optimisation, reinforcement learning, bandit algorithms, active learning, network routing.

Under the supervision of Dr. habil. Zwi Altman, Dr. Éric Gourdin (Orange Labs), and Prof. Richard Combes (CentraleSupélec, Paris-Saclay university).

Some machine-learning algorithms are on optimisation techniques, but do not always fully exploit them. For example, combinatorial bandit algorithms tend to have exponential complexity, even when the deterministic combinatorial problem has a known polynomial-time algorithm (matching, shortest path, etc.): using more advanced tools from mathematical optimisation, **I reduce the learning time complexity to polynomial** in many useful cases for two state-of-the-art algorithms (ESCB, based on the optimism principle; OSSB, based on an information-theoretic minimum bound on the regret).

Related publications: [C2], [C3], [T4], [T7].

Open-source Julia implementation: [CombinatorialBandits.jl](#). The underlying combinatorial and nonsmooth optimisation algorithms are separately available as [Kombinator.jl](#) and [NonsmoothOptim.jl](#).

Outside my thesis, I also worked on network-routing problems, more specifically the relationships between routing under uncertainty (more specifically, oblivious routing) and fair routing.

Related publications: [A5], [P2], [P3], [T6], [T7].

Open-source Julia implementation: [Seleroute.jl](#).

Along with these topics, I contributed to RAL, a novel active-learning methodology that builds upon reinforcement learning.

Related publications: [J2], [C1], [W2], [A7].

Open-source Python implementation of the methodology: [RAL](#).

Full text: <https://theses.hal.science/tel-03296009v1>

Grants: TMA 2018 student grant, ACM SIGMETRICS 2021 student grant.

January 2016-September 2017: **research engineer** at the university of Liège, Belgium.

Keywords: mathematical optimisation, data analytics.

In partnership with N-SIDE (project coordinator), UCL (ICTEAM and CRECIS), and ICEDD, within the InduStore research project.

Industry may take advantage of the **increasing electricity price volatility**, by organising its production around price forecasts, and provide flexibility services to the grid. Moreover, respecting the well-being of the workforce is a requirement in this context. My responsibilities were to develop, apply, and evaluate mathematical **optimisation models** of plant operations, including HR concerns.

Related publications: [A2], [A3], [P1], [J3].

The result of this work is available as open-source Julia software: [IndustrialProcessFlexibilisation.jl](https://github.com/IndustrialProcessFlexibilisation/jl).

September 2015-December 2015: **research engineer** at the university of Liège, Belgium.

Keyword: mathematical optimisation, mathematical modelling.

In partnership with the Walloon region and university of Liège (HECE).

Water-reservoir (such as dams) operators usually apply fixed rules or metaheuristics. However, these solutions are not always truly optimum for the sought objectives. My responsibility was to define **new operational rules** based on historical data using mathematical optimisation.

Related publication: [J1], [A6], [T7].

The result of this work is available as open-source Julia software: [ReservoirManagement.jl](https://github.com/ReservoirManagement/jl).

September 2014-June 2015: **master's thesis** at the university of Liège, Belgium.

Keywords: mathematical optimisation, mathematical modelling under uncertainty.

The two main paradigms to model uncertainty in mathematical optimisation are **stochastic and robust programming**. For the facility-location and unit-commitment problems, I conducted an **in-depth comparison** of the cost and the robustness of the obtained solutions.

Full text: <http://hdl.handle.net/2268/197090>.

Related publications: [A5], [A1], [T3], [T7].

Award: best master's thesis in computer science and engineering, AIM, 2015.

Under the supervision of Prof. Quentin Louveaux.

July-August 2014: **intern** at N-SIDE (Louvain-la-Neuve, Belgium), working on the ENERTOP project.

Keywords: mathematical optimisation, mathematical modelling.

In the context of the **optimisation of energy costs** in a complex plant, my **responsibilities** were to make the solutions more **robust** to uncertain events, such as failures, and to consider the electricity-price uncertainty. To this end, I contributed to the Scala production code base.

Under the supervision of Prof. Bertrand Cornélusse (N-SIDE, now university of Liège).

Education

2017 – 2021: **PhD in Science and Technology of Information and Communication**, CentraleSupélec (Paris-Saclay university¹).

PhD thesis: Polynomial-time algorithms for combinatorial semibandits: computationally tractable reinforcement learning in complex environments.

2010 – 2015: **master's degree in Computer Science and Engineering²**, university of Liège (ULg), Belgium, magna cum laude (second year and master's thesis: summa cum laude).

Master's thesis: Implementing and comparing stochastic and robust programming.

Award: best master's thesis in computer science and engineering, AIM, 2015.

Electives:

- Artificial intelligence (machine learning, intelligent robotics).
- Applied mathematics (discrete and numerical optimisation, high-performance scientific computing).
- Networks (advanced networking, information theory, telecommunications).

Active contribution to the Revue des Ingénieurs (satiric play; 2013, 2016).

2004 – 2010: secondary education (CESS), Saint-Barthélemy, Liège, Belgium, magna cum laude.

Electives: mathematics, Latin, ancient Greek.

Distinctions: finalist for the Belgian round of the Olympics of Informatics (2010); top 3 contestant for the ancient Greek translation competition of the Rencontres Grecques, Institut du Sacré-Cœur de Mons, Belgium (2010). Student mentor in 2008-2009 and 2009-2010 for first-year students (12-year-old).

Communication experience

Journal articles

- [J4] *Statistically Efficient, Polynomial-Time Algorithms for Combinatorial Semi-Bandits.*
Thibaut Cuvelier, Richard Combes, Éric Gourdin.
Proceedings of the ACM on Measurement and Analysis of Computing Systems, vol. 5, no. 9, pp. 1–31, February 2021.
<https://hal.science/hal-03162127> and <https://arxiv.org/abs/2002.07258>
- [J3] *Embedding Reservoirs in Industrial Models to Exploit their Flexibility.*
Thibaut Cuvelier.
SN Applied Sciences, vol. 2, no. 12, article 2171, December 2020.
<https://hal.science/hal-03053322>
- [J2] *Adaptive and Reinforcement Learning Approaches for Online Network Monitoring and Analysis.*
Sarah Wassermann, Thibaut Cuvelier, Pavol Mulinka, Pedro Casas.
IEEE Transactions on Network and Service Management, vol. 18, no. 2, June 2021.
<https://hal.science/hal-03110834>
- [J1] *Comparison Between Robust and Stochastic Optimisation for Long-term Reservoir Operations Under Uncertainty.*
Thibaut Cuvelier, Pierre Archambeau, Benjamin Dewals, Quentin Louveaux.
Water Resources Management, vol. 32, no. 5, pp. 1599–1614, March 2018.
<http://hdl.handle.net/2268/219394> and <https://arxiv.org/abs/1801.08892>

¹ Ranked 1st in mathematics in the world in 2020 and 2021 [by Shanghai Ranking Consultancy](#).

² [The French Commission des Titres d'Ingénieur \(CTI\)](#) awarded this master's degree [a EUR-ACE label](#).

Conference articles

- [C4] *Generative Large Neighborhood Search: Scalable Set Cover Optimization via Discrete Diffusion*
Achref Jaziri, Thibaut Cuvelier, Bruno De Backer.
International Conference on Machine Learning (ICML), Seoul (South Korea), August 2026.
<https://icml.cc/virtual/2026/poster/60514>
- [C3] *Asymptotically Optimal Strategies for Combinatorial Semi-Bandits in Polynomial Time.*
Thibaut Cuvelier, Richard Combes, Éric Gourdin.
Algorithmic Learning Theory (ALT), Paris (France), March 2021.
29% acceptance rate.
<https://hal.science/hal-03162140> and <https://arxiv.org/abs/2102.07254>
- [C2] *Statistically Efficient, Polynomial-Time Algorithms for Combinatorial Semi-Bandits.*
Thibaut Cuvelier, Richard Combes, Éric Gourdin.
ACM SIGMETRICS 2021, Beijing (China) and online, June 2021.
12% acceptance rate.
<https://hal.science/hal-03201526>
- [C1] *ADAM & RAL: Adaptive Memory Learning and Reinforcement Active Learning for Network Monitoring.*
Sarah Wassermann, Thibaut Cuvelier, Pedro Casas, Pavol Mulinka.
15th International Conference on Network and Service Management (CNSM) 2019, Halifax (Nova Scotia, Canada), October 2019.
16% acceptance rate.
<https://hal.science/hal-02301393>

Workshop articles

- [W3] *Middle-Mile Logistics Through the Lens of Goal-Conditioned Reinforcement Learning*
Onno Eberhard, Thibaut Cuvelier, Michal Valko, Bruno de Backer
Conference on Neural Information Processing Systems (NeurIPS) Goal-Conditioned Reinforcement Learning Workshop, New Orleans (USA), December 2023.
- [W2] *RAL: Improving Stream-Based Active Learning by Reinforcement Learning.*
Sarah Wassermann, Thibaut Cuvelier, Pedro Casas.
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD) Workshop on Interactive Adaptive Learning (IAL), Würzburg (Germany), September 2019.
<https://hal.science/hal-02265426/>
- [W1] *NETPerfTrace — Predicting Internet Path Dynamics and Performance with Machine Learning.*
Sarah Wassermann, Pedro Casas, Thibaut Cuvelier, Benoît Donnet.
ACM SIGCOMM Workshop on Big Data Analytics and Machine Learning for Data Communication (Big-DAMA), Los Angeles (California, USA), August 2017.
<http://hdl.handle.net/2268/211667>

Abstracts

- [A14] *ORTools.jl: access Google's solvers through JuMP.*
Warren Ochibobo, Thibaut Cuvelier.
JuliaCon Local Paris 2025, Paris (France), October 2025.
- [A13] *ORTools.jl: access Google's solvers through JuMP.*
Warren Ochibobo, Thibaut Cuvelier.
JuliaCon 2025, Pittsburgh (Pennsylvania, USA), July 2025.

- [A12] *Getting to 1,000x: Solving Large Set-Covering Problems.*
Bruno De Backer, Thibaut Cuvelier.
Optimization Days JOPT 2025, Montréal (Québec, Canada), May 2025.
<https://hal.science/hal-05068198>
- [A11] *A Novel Instance Generator for Simulating Middle-Mile Logistics Networks.*
Aymane Lotfi, Matteo Petris, Bruno De Backer, Thibaut Cuvelier, Claudia Archetti.
33rd European Conference on Operational Research 2024, Copenhagen (Denmark), July 2024.
<https://hal.science/hal-04755189v1>
- [A10] *A Tale of Middle-Mile Logistics, Graph Neural Networks, and Reinforcement Learning.*
Onno Eberhard, Thibaut Cuvelier, Bruno De Backer.
33rd European Conference on Operational Research 2024, Copenhagen (Denmark), July 2024.
<https://hal.science/hal-04755188v1>
- [A9] *OR-Tools' Vehicle Routing Solver: a Generic Constraint-Programming Solver with Heuristic Search for Routing Problems.*
Thibaut Cuvelier, Frédéric Didier, Vincent Furnon, Steven Gay, Sarah Mohajeri, Laurent Perron.
Congrès de la Société française de recherche opérationnelle et d'aide à la décision (ROADEF) 2023, Rennes (France), February 2023.
https://tcuvelier.be/files/ortools_roadef2023.pdf
- [A8] *ConstraintProgrammingExtensions: MathOptInterface gets broader usage.*
Thibaut Cuvelier.
JuMP-dev 2021, online, July 2021.
<https://hal.science/hal-03306211v1>
- [A7] *Improving Stream-Based Active Learning with Reinforcement Learning.*
Sarah Wassermann, Thibaut Cuvelier, Pedro Casas.
Workshop for Women in Machine Learning (WiML) 2019, Vancouver (British Columbia, Canada), December 2019.
<https://hal.science/hal-02375296>
- [A6] *Operation rules of the Vesdre reservoir revisited.*
Benjamin Dewals, Thibaut Cuvelier, Pierre Archambeau, Sébastien Erpicum, Michel Piroton, Quentin Louveaux.
6th International Symposium on Hydrological Modelling of the Meuse basin, Liège (Belgium), September 2019.
<http://hdl.handle.net/2268/239415>
- [A5] *Comparing Oblivious and Robust Routing Approaches.*
Thibaut Cuvelier and Éric Gourdin.
Programme Gaspard Monge pour l'optimisation, la recherche opérationnelle et leurs interactions avec les sciences des données (PGMO Days) 2018, Palaiseau (France), November 2018.
<http://hdl.handle.net/2268/229784>
- [A4] *Retour d'expérience sur Julia pour la recherche et l'enseignement en recherche opérationnelle.*
Thibaut Cuvelier.
Congrès de la Société française de recherche opérationnelle et d'aide à la décision (ROADEF) 2018, Lorient (France), February 2018.
<http://hdl.handle.net/2268/220267>
- [A3] *Optimising workforce and energy costs by exploiting production flexibility.*
Thibaut Cuvelier and Quentin Louveaux.
21st Conference of the International Federation of Operational Research Societies (IFORS), Québec (Québec, Canada), July 2017.
<http://hdl.handle.net/2268/207330>

- [A2] *Modelling the industrial flexibility from the electricity consumption and HR points of view.*
Thibaut Cuvelier and Quentin Louveaux.
22nd Belgian Mathematical Optimization Workshop, COMEX (combinatorial optimisation: metaheuristics and exact methods), La Roche-en-Ardenne (Belgium), April 2017.
<http://hdl.handle.net/2268/209469>
- [A1] *Optimisation and uncertainty: comparing stochastic and robust programming.*
Thibaut Cuvelier.
30th Annual Conference of the Belgian Operational Research Society (ORBEL), Louvain-la-Neuve (Belgium), January 2016.
<http://hdl.handle.net/2268/197081>

Posters

- [P5] *Flattening the learning curse of powerful APIs: the use-case of vehicle routing.*
Thibaut Cuvelier, Mark Daly, Anne-Claire Hauray.
Google Research conference, Mountain View (California, USA), April 2025. (Internal.)
- [P4] *RAL — Reinforcement Active Learning for Network Traffic Monitoring and Analysis.*
Sarah Wassermann, Thibaut Cuvelier, Pedro Casas.
Proceedings of the ACM SIGCOMM Conference Posters and Demos, online, August 2020.
<https://hal.science/hal-02932839>
- [P3] *Oblivious Routing: Static Routing Prepared Against Network Traffic and Link Failures.*
Thibaut Cuvelier and Éric Gourdin.
Network Traffic Measurement and Analysis (TMA) PhD School 2019, Paris (France), June 2019.
<https://hal.science/hal-02161708/>
- [P2] *Oblivious Routing: Worst-Case Routing is not Breaking the Internet's Legs.*
Thibaut Cuvelier.
Network Traffic Measurement and Analysis (TMA) PhD School 2018, Vienna (Austria), June 2018.
<http://hdl.handle.net/2268/227128>
- [P1] *Characterising Industrial Sites' Flexibility with Reservoir Models.*
Thibaut Cuvelier.
DS3 Data Science Summer School (École Polytechnique), Paris (France), August 2017.
<http://hdl.handle.net/2268/212703>

Books

- [B3] *Créer des applications graphiques en Python avec PyQt5*, D-BookeR, March 2017.
Thibaut Cuvelier, Pierre Denis.
ISBN-13: 978-2-8227-0518-9.
- [B2] *Créer des applications avec Qt 5 – les essentiels*, D-BookeR, November 2013.
Guillaume Belz, Thibaut Cuvelier, Ilya Diallo, Louis du Verdier, Vincent Meyer, Florent Renault.
ISBN-13: 978-2-8227-0108-2.
- [B1] *Web sémantique : méthodes et outils pour le Web de données*, Pearson, May 2012.
Translated by Thibaut Cuvelier, Julien Plu, Antoine Seilles.
Original title: *Linked Data: Evolving the Web into a Global Data Space.*
Tom Heath and Christian Bizer.
ISBN-13: 978-2-7440-2519-8.

Talks

- [T9] *Unlocking the Power of Google OR-Tools with MathOptInterface.jl.*
Warren Ochibobo, Thibaut Cuvelier.
JuMP-dev workshop 2025, Auckland (New Zealand), November 2025.

- [T8] *Applications of reinforcement learning to combinatorial optimisation*, Google + CNRS, Paris (France), May 2024.
https://tcuvelier.be/files/rl_or_cnrs2024.pdf
- [T7] *A Journey through Uncertain Optimisation*, Amazon Transport Services, Luxembourg (Luxembourg), November 2023.
https://tcuvelier.be/files/uncertainty_ats2023.pdf
- [T6] *Seleroute.jl, a generic package for network-routing optimization*, Julia and Optimization Days 2023, Paris (France), October 2023.
<https://hal.science/hal-04231900>
- [T5] *À la découverte de Julia !*, Datacraft, Paris (France), November 2022.
<https://hal.science/hal-03870998>
<https://www.youtube.com/watch?v=rSrhKkk70M0>
- [T4] *Polynomial-Time Combinatorial Bandits: Computationally Tractable Reinforcement Learning in Complex Environments*, Amazon Transport Services, Luxembourg (Luxembourg), March 2021.
https://tcuvelier.be/files/bandits_ats2021.pdf
- [T3] *Voyage incertain : découvrir l'optimisation stochastique et robuste*, Orange Labs, Châtillon (France), February 2018.
<http://hdl.handle.net/2268/219824>
- [T2] *A Journey through Julia*, IEEE Student Branch Liège (Belgium), May 2017.
<http://hdl.handle.net/2268/210211>
- [T1] *A Journey through Julia*, Geeks anonymes, Liège (Belgium), November 2016.
<http://hdl.handle.net/2268/203491>

Invited talk

- [I2] *OR-Tools' Vehicle Routing Solver: a Generic Constraint-Programming Solver with Heuristic Search for Routing Problems.*
Thibaut Cuvelier, Frederic Didier, Vincent Furnon, Steven Gay, Sarah Mohajeri, Laurent Perron.
Datacraft 2026, Paris (France), May 2026.
- [I1] *Bridging Continuous and Discrete Optimization in Julia.*
Thibaut Cuvelier.
ICCOPT 2025, Los Angeles (California, USA), July 2025.

Open-source contributions

Research codes:

- Maintainer of [Google OR-Tools](#), while working at Google (since 2021), [A9], [A12], [A13], [A14], [T13], [I1]. C++.
- Main developer of [CombinatorialBandits.jl](#) (2020-2021), [C2], [C3]. Julia.
- Main developer of [Seleroute.jl](#) (computer-network routing, 2020-2023), [A5], [P2], [P3]. Julia.
- Main developer of [IndustrialProcessFlexibilisation.jl](#) (flexible industrial operations, 2017-2018), [A2], [A3], [P1], [J3]. Julia.
- Main developer of [ReservoirManagement.jl](#) (water-reservoir operations, 2016-2018), [J1], [A6]. Julia.
- Co-developer of [RAL](#) (for stream-based active learning, 2019-2020), [J2], [C1], [W2], [A7]. Python.

Libraries used for research:

- Main developer of [NonsmoothOptim.jl](#), a nonsmooth-optimisation Julia package (2020-2021).
- Main developer of [Kombinator.jl](#), a combinatorial-optimisation Julia package (2020-2021).

- Main developer of a generic interface for constraint programming in Julia [A8], [I1], [highlighted in an official blog post: ConstraintProgrammingExtensions.jl](#) (2020-2021), [JuCP.jl](#) (2020-2021), [CPLEXCP.jl](#) (2020-2021), [Chuffed.jl](#) (2021), [JaCoP.jl](#) (2021).
- Regular contributor to several optimisation-oriented Julia packages (access to infeasibility information from the solvers, export to LP format): [JuMP.jl](#) (2019-2020), [MathOptInterface.jl](#) (2019-2020), [MathOptFormat.jl](#) (2019, since then merged with the latter), [MathOptFormat](#). Developer of the [CPLEX.jl](#) optimisation-solver wrapper (since 2021; contributor in 2019-2021). Regular contributor to several optimisation-solver wrappers: [Gurobi.jl](#) (2019-2021), [SCS.jl](#) (2019), [Xpress.jl](#) (2019).
- Occasional contributor to several Julia packages: [TimeSeries.jl](#) (2015-2016), [Distributions.jl](#) (2016), [Nemo.jl](#) (2017), [Hungarian.jl](#) (2018, 2022), [LightGraphsMatching.jl](#) (2018, now [GraphsMatching.jl](#)), [SimpleWeightedGraphs.jl](#) (2018), [JavaCall.jl](#) (2020-2021).

Documentation and technical-writing tools:

- Main developer of [QtDocTools](#) (Java/XSLT, since 2014; management of the translation of Qt's documentation, including tooling to work with DocBook documents).
- Developer of [LyX](#) (C++/Python, since 2018; maintainer of DocBook, ePub, XHTML, and MathML output formats).
- Occasional contributor to [Apache POI](#) (Java, 2019-2020).
- Occasional contributor to [Qt](#) (C++, 2019-2025; export from QDoc to DocBook), [120+ patches merged. Recognised as a Top Contributor in 2022.](#)
- Occasional contributor to [LilyPond](#) (Python, [2021](#), DocBook fixes).
- Occasional external contributor to the [DocBook](#) OASIS standard ([2016-2022](#)).

Miscellaneous (related to my other activities on Developpez.com):

- Contribution to the translation into French of [Qt Creator](#) ([2011-2013](#)).
- Occasional contributor to PHP libraries: [Silex](#) ([2012](#)), [GeSHi](#) ([2014](#)).

Service to the community

Reviewer for the following journals:

- Journal of Hydrology (2019-2025).
- Journal of Water Resources Management (2019-2026).
- International Journal of Mining Science and Technology (2021).
- Operations Research Perspectives (2021).
- Journal of Open Source Software (2025).
- Transactions on Machine Learning Research (2025-2026).

Reviewer for the following conferences:

- International Conference on Water Resource and Environment conferences: 7th (WRE 2021), 10th (WRE 2024), 11th (WRE 2025).
- JuliaCon: 2019, 2024, 2025.
- 16th International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt 2018).

Session co-chair at IFORS 2017.

Trainings

First aid: European first aid certificate (BEPS, Belgian Red Cross, May 2015); workplace first aider (UNASS, December 2018 and February 2020; Pulsart, December 2023).

Graduate school trainings:

- STIC: law and intellectual property; European projects Horizon 2020; ethics and technology (2019).
- SOCN: [algorithmic convex optimisation](#) (2016).

PhD schools:

- [HIM School on modern directions in discrete optimization 2021](#).
- [Combinatorial Optimization at Work \(CO@W\) 2020](#).
- [TMA 2019 PhD school](#).
- [ECML PKDD 2019 Summer School](#), quality-of-experience track.
- 2018 school on column generation.
- [TMA 2018 PhD school](#).
- IPCO 2016 summer school.

Other trainings at university of Liège: open-channel flows ([GCIV2034](#), 2015) and fluvial hydrodynamics ([GCIV2035](#), 2016).

Pedagogical training

Ten courses at the [Institut de Formation et de Recherche en Enseignement Supérieur](#) (IFRES, university of Liège) — equivalent to the [FORMASTART](#) certificate:

- Design multimedia material for face-to-face teaching.
- Competency-based approach.
- Motivate my students.
- Introduction to evaluation: principles and quality criteria.
- Customising teaching by considering students' characteristics.
- Triple concordance between objectives, methods, and evaluation.
- Organising practical lessons in science and applied science courses.
- Introduction to the Blackboard Learn platform.
- Using Blackboard Learn to create tests.
- Evaluate and regulate one's teaching activities.

Online courses

Machine learning:

- [Deep learning specialisation, Andrew Ng, Coursera, March 2018 \(overall score: 100%\)](#).
- Reinforcement Learning Explained, Microsoft DAT257x, edX.
- [Artificial Intelligence A-Z™: Learn How To Build An AI, SuperDataScience, Udemy, April 2020](#).
- [Advanced AI: Deep Reinforcement Learning in Python, Lazy Programmer, Udemy, August 2020](#).
- [Unsupervised Deep Learning in Python, Lazy Programmer, Udemy, May 2021](#).
- [Understanding Artificial Intelligence through Algorithmic Information Theory, edX, July 2021](#).

Networks and telecommunications:

- Comprendre la 4G, IMT Atlantique, Fun MOOC, December 2020.
- Explorer la 5G, IMT Atlantique, Fun MOOC, May 2021.
- Comprendre le cœur d'internet : les réseaux d'opérateurs, IMT Atlantique, Fun MOOC, December 2021.
- Les réseaux d'accès optiques FTTH, Mines-Télécom, Fun MOOC, May 2022.

Computer skills:

- [Linux Administration with Troubleshooting Skills: Hands-On, Udemy, December 2022](#).
- [Advanced Linux System Administration, Udemy, December 2022](#).

Pedagogy:

- [Leaders of Learning, HarvardX, edX, September 2021](#).
- Se former pour enseigner dans le supérieur, ministère de l'Enseignement supérieur (France), Fun MOOC, December 2021.
- Psychologie pour les enseignants, Paris Sciences & Lettres, Fun MOOC, December 2021.
- Transformer l'enseignement et la formation : mission hybridation, Cergy Paris Université, Fun MOOC, July 2022.

Non-research work experience

- 2018–now: book-review manager for **Developpez.com**. Partnership management and development (7 new publishers).
- August 2015: development of a real-time car-sharing prototype application for the **ULg**. This prototype evolved into the UGo platform (<https://ugo.be/>).
Technologies: Python, Django.
Under the supervision of Prof. Bertrand Cornélusse (ULg).
- July 2015: translation into English of a 100-page numerical-analysis textbook (2nd year students) for the **ULg**.
Under the supervision of Prof. Quentin Louveaux (ULg).
- 2011–2017: officer for the ULg **IEEE Student Branch**. Tasks: event organisation, website and server maintenance, poster design.
Technologies: Joomla!, CentOS.
- 2009–now: section manager for **Developpez.com** (Qt since 2009, project hosting in 2010 and 2011, semantic Web between 2011 and 2013, HPC since its creation in 2018, algorithms and mathematics since 2018). Tasks: team management, website maintenance, application development.
- 2008–now: author (30+ articles), translator (100+ articles), proofreader, columnist (750+ news articles), and technical book critic (145+ books) for **Developpez.com**.
<http://tcuvelier.developpez.com/> and <https://tcuvelier.wordpress.com/>

Pedagogical experience

- Spring 2019: exercise sessions for the **combinatorial optimisation** course (master's students), in French, with a 20-student class. University of Paris-Sud (now Paris-Saclay university).
Lecturer: Prof. Abdel Lisser.
- Winter 2018: instructor for the introduction to **object-oriented programming and Java** (bachelor's students), in French, with a 20-student class. University of Paris-Sud (now Paris-Saclay university).
Lecturer: Prof. Guillaume Wisniewski.
- Winter 2018: exercise sessions for the introduction to **probabilities** course (bachelor's students), in French, with a 30-student class. University of Paris-Sud (now Paris-Saclay university).
Lecturer: Prof. Abdel Lisser.
- Spring 2017: project supervision for the **intelligent robotics** course (master's students), in English, with a 50-student class. University of Liège.
Lecturers: Prof. Bernard Boigelot, Philippe Latour, Antoine Lejeune, Dr. Raphaël Marée, Prof. Marc Van Droogenbroeck, Prof. Louis Wehenkel.

- Winter 2016: exercise sessions and project supervision for the **discrete optimisation** course (master's students), in English, with a 50-student class. Includes the design of a complete exercise book. University of Liège.
Lecturer: Prof. Quentin Louveaux.
- Spring 2016: project supervision for the **intelligent robotics** course (master's students), in English, with a 50-student class. University of Liège.
Lecturer: Dr. Renaud Detry.
- Winter 2015: exercise sessions and project supervision for the **discrete optimisation** course (master's students), in English, with a 50-student class. University of Liège.
Lecturer: Prof. Quentin Louveaux.
- Spring 2015: student instructor for the **numerical analysis project** (first-year students), in French, with a 20-student class. University of Liège.
Lecturer: Prof. Quentin Louveaux.

Open course material

Discrete optimisation: development of an exercise book with solutions and Julia/JuMP source-code examples. 2016-2022. Available at <https://github.com/dourouc05/OptimisationTeachingKit>.

Intelligent robotics: contributions to the TRS project (*teaching robotics with a simulator*), including updates of the webpages and video making. 2017-2020. Available at:
<https://github.com/ULgRobotics/trs>. Official website: <http://ulgrobotics.github.io/trs/>.

Language skills

French	Native
English	Cambridge FCE (B2 level) in 2010, English-taught MSc (2013-2015), British Council EnglishScore (C1) in 2020
German	B1-level training from 2014 to 2017

Technical skills

Programming

Programming languages	Python, C++, Julia, Java, C, PHP, Scala
Mathematical modelling	JuMP (Julia), AMPL
Optimisation solvers	Gurobi, CPLEX, Bonmin, Couenne
Data analytics	scikit-learn
Data visualisation	Mathematica, Plotly and Dash, MATLAB
Query languages	SQL, SPARQL
Database systems	Microsoft SQL Server, MySQL, SQLite
XML technologies	XSLT, XPath, RELAX NG, XSpec
Development environments	PyCharm (Python), IntelliJ IDEA (Java), CLion and Visual Studio (C++), Visual Studio Code (Julia), MATLAB, Mathematica
Graphical user interfaces	Qt (C++) and PyQt (Python), especially Qt Quick
Version control	Mercurial (Hg), Git, SVN

Office software

Word processors	LyX (LaTeX), Microsoft Word
Spreadsheet	Microsoft Excel
Technical documentation	DocBook
Technical-writing tools	Oxygen XML Author, XMLmind
Technical drawing	Microsoft Visio, Adobe Illustrator
Photo editing	Adobe Photoshop

Personality

Detail-minded, results-driven, autonomous, energised by challenges.

Sports: climbing (since 2002, both indoor and outdoor), walking.

Hobbies: reading novels (thrillers); listening to music (progressive rock, electronica).